

Web Development Capstone Project Presentation

A Modern, Concurrency-Safe Full-Stack MERN Event Ticketing Platform

SKILLORBIT: EVENT

PRESENTED BY

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LIVE DEPLOYED APPLICATION:

<https://skillorbit-event-booking-system.vercel.app/>

WHY EVENT BOOKING SYSTEMS FAIL

Operational Vulnerabilities in Traditional Systems

- **High-Concurrency Race Conditions:** Simultaneous booking requests during peak ticket drops cause overbooking beyond actual venue capacity.
- **Seat Hoarding & Lack of Bounds:** Absence of per-user reservation limits allows individuals or automated scripts to hoard entire blocks of tickets.
- **Opaque Attendance Data:** Organizers lack real-time visibility into registration velocity, category distribution, and seating utilization.
- **Manual Spreadsheet Chaos:** Campus and corporate symposiums rely on error-prone Google Sheets or paper rosters for manual gate check-in.
- **Lack of Immediate Digital Pass:** Attendees receive delayed or fragmented confirmation without printable admission credentials.

The Core Engineering Question

"How can we engineer a unified, decoupled full-stack platform that guarantees atomic concurrency control, fair inventory allocation, and administrative analytics without database thread-locking or write starvation?"

Key System Requirements:

- ✓ **Atomic Seat Decrements:** Execute conditional updates so capacity never drops below 0.
- ✓ **Fair Allocation Limits:** Strict 10-ticket reservation ceiling per attendee per event.
- ✓ **Stateless Role Security:** JWT Bearer tokens protecting private attendee and admin APIs.
- ✓ **Instant Reporting:** Streaming RFC 4180 CSV generation for check-in audits.

Project Objectives

SIX CORE TECHNICAL PILLARS

1. Zero Overselling Guarantee

Implement atomic database-level seat decrementing using MongoDB conditional operations (\$gte) to prevent negative capacity and race conditions under peak concurrent loads.

2. Fair Allocation & Anti-Hoarding

Enforce strict reservation bounds capping individual attendee bookings at a maximum of 10 tickets per event, ensuring equitable campus event access.

3. Stateless Multi-Role Security

Architect a robust Role-Based Access Control (RBAC) model using JSON Web Tokens (JWT) and Bcrypt password hashing, segregating attendee routes from admin controls.

4. Instant Digital Boarding Passes

Generate unique, collision-resistant reference codes (EVT-YYYYMMDD-XXXX) and modal admission passes with QR stub simulation and printer stylesheet formatting.

5. Executive Visual Analytics

Embed an interactive telemetry engine using Recharts (Area and Donut charts) alongside one-click RFC 4180 CSV exports for physical check-in verification.

6. 100% Automated Test Verification

Validate all functional workflows and defensive edge cases using a comprehensive automated test runner, ensuring zero defects across 77 integration test cases.

Proposed Solution: SkillOrbit

Attendee Discovery & Ticketing Experience

- **Real-Time Event Discovery:** Multi-category filtering across conferences, workshops, and symposiums with debounced text search.
- **Instant Seat Decrementing:** Atomic reservations execute immediately without thread locking or stale inventory.
- **Digital Admission Passes:** Formatted pass modal complete with attendee details, reference codes, and print layout.
- **Self-Service Portal:** Personalized dashboard displaying upcoming schedule, booking history, and active tickets.
- **Zero-Friction Cancellation:** Attendees can cancel bookings with instant, automated restitution of seats back to the event pool.

Organizer Governance & Analytics Engine

- **Full Event CRUD Lifecycle:** Create, update, draft, and publish events with dual-mode banner posters (local file uploads or remote URLs).
- **Live Business Intelligence:** Real-time KPI metrics on total bookings, capacity utilization %, revenue, and user accounts.
- **Visual Telemetry (Recharts):** AreaChart for monthly reservation trends and Donut PieChart for category distribution.
- **Streaming CSV Exports:** One-click download of RFC 4180 compliant attendee rosters and venue capacity audits.
- **Audit Trail Preservation:** Events with confirmed bookings are safely transitioned to cancelled status to prevent data loss.

COMPREHENSIVE CAPSTONE IMPLEMENTATION

Key Platform Features

User Authentication & Sessions

Bcrypt salted hashing (10 rounds), stateless Bearer JWT tokens, and automated Axios interceptors with persistent LocalStorage state.

Concurrency-Safe Reservations

Atomic conditional decrementing (\$gte query + \$inc) prevents double-booking and negative capacity under concurrent access.

Administrative Command Center

Unified management console featuring event lifecycle toggles, protected capacity editing, and user account oversight.

Real-Time Inventory Badges

Dynamic status pills indicating 'Sold Out', 'Selling Fast' (<20% seats remaining), or high availability calculated directly from live database counts.

Personalized Attendee Dashboard

Real-time counters for active passes, upcoming event schedules, booking history, and dynamic event recommendations.

Automated CSV Reporting Engine

Server-side streaming generation of flattened RFC 4180 CSV files for physical venue gate check-in and capacity audits.

Technology Stack

PRODUCTION-GRADE LIBRARIES & FRAMEWORKS

Architectural Layer	Technology / Framework	Version	Role & Engineering Responsibility
Frontend Client	React 19 + Vite 8	19.0.0 / 8.3.1	Single Page Application with fast HMR and sub-second builds
UI Styling & Icons	Tailwind CSS + Lucide React	3.4.17 / 1.16.0	Responsive styling, high-contrast system colors, accessible icons
Data Visualization	Recharts Telemetry	3.7.0	SVG AreaCharts for booking trends and PieCharts for categories
Backend REST API	Node.js + Express.js	v20 LTS / 4.21.2	Asynchronous HTTP routing, middleware, controllers, error handling
Database & ODM	MongoDB Atlas + Mongoose	Cloud M0 / 8.9.5	Document storage, atomic operations, compound & text indexing
Authentication	JWT + BcryptJS	9.0.2 / 2.4.3	Stateless HTTP Bearer token authorization and salted hashing
Media & Reporting	Multer + json2csv	1.4.5 / 6.0.0	Local event banner uploads and RFC 4180 streaming CSV generation
Testing Suite	Native Node.js Test Runner	Node v20 Test	77-case automated E2E integration verification suite (npm test)

System Architecture

DECOUPLED MULTI-TIER ARCHITECTURE

Tier 1: Client Presentation Layer

React 19 Single Page Application bundled with Vite 8. Implements responsive Tailwind CSS styling, AuthContext and ToastContext providers, and protected route wrappers.

Tier 2: Network & API Gateway Layer

RESTful HTTP/JSON endpoints over HTTPS. Centralized Axios API services with request/response interceptors automatically injecting Bearer JWT headers.

Tier 3: Business Logic & Middleware Layer

Express 4.21 application server running on Node.js v20. Modular routers protect endpoints with JWT authentication and strict role-based authorization (admin vs user).

Tier 4: Data Persistence & Cloud Tier

MongoDB Atlas managed cloud replica set. Mongoose 8.9 ODM handles schema enforcement, compound query indexing, text search, and concurrency-safe atomic operations.

Database Design & Data Modeling

NORMALIZED RELATIONAL DOCUMENT MODEL

Entity / Collection	Key Schema Fields	Data Types & Constraints	Relationships & Indexing Strategy
Users Collection	name, email, password, role, createdAt	email: Unique, Lowercase, Regex password: Hashed (select: false) role: enum ['user', 'admin']	1-to-Many with Bookings. Unique Index on email. Sub-10ms lookup latency.
Events Collection	title, description, date, venue, category, capacity, availableSeats, price, status, image, createdBy	capacity: Number (min: 1) availableSeats: Number (min: 0) status: ['published', 'draft', 'cancelled'] createdBy: ObjectId (Ref: User)	1-to-Many with Bookings. Compound Index: { date: 1, status: 1 }. Full-text Search: { title, desc, venue }.
Bookings Collection	bookingReference, user, event, seatsBooked, totalPrice, status, createdAt	bookingReference: Unique String seatsBooked: Number (1 to 10) status: ['confirmed',	Many-to-1 with Users & Events. Compound Index: { user: 1, createdAt: -1 }. Atomic capacity rollback

User Authentication &

Authentication Architecture

- **Input Validation:** Strict verification of name, email format, and password length (minimum 6 characters) on registration.
- **Cryptographic Hashing:** Passwords encrypted with bcryptjs using 10 salt rounds; excluded by default from query projections.
- **Signed JWT Tokens:** Stateless JSON Web Tokens signed with HMAC-SHA256, carrying user ID and role with 7-day expiration.
- **Token Interceptor:** Axios request interceptors inject 'Bearer <token>' headers on all authenticated API requests.
- **Session Resilience:** Automatic session restoration via LocalStorage and /api/auth/me profile verification.

Role-Based Access Control (RBAC)

- **Granular Route Gates:** Express auth middleware verifies JWTs; authorize('admin') middleware restricts privileged operations.
- **Attendee Privileges:** Authenticated users can browse events, book up to 10 tickets, view digital passes, and cancel their own bookings.
- **Admin Privileges:** Administrators have exclusive access to event CRUD, capacity editing, platform telemetry, and CSV report downloads.
- **403 Forbidden Protection:** Unauthorized role access attempts return HTTP 403 Forbidden without exposing server details.
- **Client-Side Route Guards:** React Router protected routes redirect guests to /login and prevent attendee access to /admin/*.

Event Management &

Event CRUD & Lifecycle Management

- **Full CRUD Operations:** Organizers create, preview, edit, publish, and delete events through a responsive management console.
- **Lifecycle States:** State machine controlling published (publicly visible), draft (staged work), and cancelled states.
- **Dual-Mode Media Banners:** Supports both direct local file image uploads via Multer and remote high-resolution image URLs.
- **Capacity Protection:** Validation guards prevent reducing event capacity below the number of already confirmed tickets.
- **Rich Event Metadata:** Structured capture of event title, schedule, venue location, category, ticket price, and descriptions.

Safe Deletion & Audit Trail Protection

- **The Audit Risk:** Hard-deleting events with confirmed attendee reservations destroys historical financial and booking records.
- **Automatic State Transition:** When an admin attempts to delete an event with active bookings, the system transitions it to 'cancelled' status.
- **Safe Removal of Empty Events:** Unpublished drafts or concluded events with zero confirmed attendees are safely purged from the database.
- **Referential Integrity:** Guarantees zero orphaned booking documents and maintains historical reporting integrity.
- **Attendee Notification State:** Cancelled status is immediately reflected across all attendee dashboards and passes.

Event Discovery & Real-

Dynamic Search & Multi-Filter Engine

- **Category Filter Bar:** Filter events by domain: All, Conference, Workshop, Concert, College Fest, Hackathon, or Symposium.
- **Debounced Keyword Search:** Instant 250ms debounced search querying across event titles, venue locations, and descriptions.
- **Chronological Scheduling:** Events sorted chronologically; past events are automatically flagged or excluded from public discovery.
- **Zero-State Graceful Fallback:** Clean empty state views with an intuitive 'Clear All Filters' action when queries yield zero records.
- **Mobile-Optimized Cards:** Responsive grid layout (1 col mobile, 2 col tablet, 3 col desktop) with zero UI layout shift.

Live Inventory & Availability Indicators

- **Real-Time Capacity Counters:** Displays exact available seats vs total venue capacity (e.g., '17 of 20 seats remaining').
- **Sold Out Status Guard:** Immediate 'Sold Out' badge with disabled action buttons when availableSeats drops to 0.
- **Selling Fast Indicator:** Amber warning banner activates when remaining capacity falls below 20%, encouraging early booking.
- **Visual Progress Fill:** Clean progress bar showing percentage of venue capacity currently occupied.
- **Event Status Badges:** Distinct status indicators differentiating published, concluded, and cancelled events.

Ticket Booking Workflow

End-to-End Booking Lifecycle

Step 1 – Event Selection: Attendee navigates the catalog, reviews event details, and initiates ticket booking.

Step 2 – Seat Quantity: User selects ticket count (1 to 10 seats) with live pricing calculation in an accessible modal.

Step 3 – Atomic Decrement: Client dispatches POST /api/bookings; backend executes conditional atomic seat update.

Step 4 – Reference Generation: System generates a unique, human-readable code: EVT-YYYYMMDD-XXXX.

Step 5 – Pass Presentation: Instant digital boarding pass rendered on-screen with QR stub simulation and print view.

Digital Boarding Pass Architecture

- **Unique Reference Identifier:** Collision-resistant formatted code (e.g., EVT-20261004-C323DE) enabling rapid lookups.
- **Verified Attendee Credentials:** Displays authenticated attendee name, registered email, and timestamped transaction log.
- **Venue & Schedule Details:** Shows event venue, scheduled calendar date, and confirmed ticket count.
- **Simulated Gate Check-In QR Stub:** Visual digital barcode stub facilitating rapid physical gatekeeper validation.
- **Printable CSS Layout:** Includes window.print() stylesheet optimization for paper or PDF ticket issuance.

Concurrency Control & Seat Integrity

ELIMINATING RACE CONDITIONS AT THE DATABASE LAYER

MongoDB Atomic Conditional Decrement Implementation:

```
const event = await Event.findOneAndUpdate(  
  { _id: eventId, status: 'published', availableSeats: { $gte: requestedSeats } },  
  { $inc: { availableSeats: -requestedSeats } },  
  { new: true }  
);
```

Guarantees: If availableSeats < requestedSeats, the operation atomically matches 0 documents and returns null without decrementing.

Why Naive Read-Then-Write Approaches Fail:

X Thread Interleaving: Two users read availableSeats = 1 at the same millisecond.

X Race Condition: Both proceed to decrement and confirm bookings.

X Overbooking Deficit: Capacity drops to -1, creating a physical seating deficit.

The SkillOrbit Concurrency Guarantee:

✓ Document-Level Atomicity: Evaluated directly at the database engine level.

✓ Single Winner: Exactly one concurrent request succeeds; others receive clean 400 errors.

✓ Automatic Rollback: Booking cancellations execute \$inc: { availableSeats: +N }.

Attendee Dashboard &

Personalized Metrics & Schedules

- **Total Reservations KPI:** Live counter calculating all event reservations created by the authenticated user.
- **Active Passes Metric:** Real-time count of confirmed upcoming admissions ready for physical gate check-in.
- **Cancelled Reservations Log:** Transparent accounting of cancelled bookings for audit history.
- **Upcoming Schedules Section:** Chronological listing of confirmed upcoming events with dates, times, and venues.
- **Discovery Recommendations:** Dynamic sidebar suggesting newly scheduled campus symposiums and tech workshops.

Self-Service Booking Management

- **My Bookings History:** Comprehensive tabular record showing reference IDs, event names, seat counts, and status pills.
- **On-Demand Ticket Viewing:** One-click 'View Pass' action redisplays the digital modal pass complete with printable view.
- **One-Click Self-Cancellation:** Attendees can independently cancel reservations without contacting event staff.
- **Atomic Seat Restitution:** Cancellation immediately releases held seats back into the event inventory for peers.
- **Defensive Cancellation Guard:** Validation prevents cancelling past events or already cancelled bookings.

Admin Dashboard &

REAL-TIME TELEMETRY WITH RECHARTS

Executive KPI Metrics Suite

- **Total Scheduled Events:** Aggregated inventory of published and draft symposiums across all categories.
- **Total Tickets Booked:** Cumulative seat reservations across the platform calculated directly from active bookings.
- **Capacity Utilization Rate:** Percentage metric calculating booked seats against total venue capacity across all events.
- **Registered User Accounts:** Total registered attendee and administrative accounts in the database.
- **Simulated Revenue Metric:** Financial summation calculated from confirmed ticket volume and ticket pricing tiers.

Interactive Recharts Visualizations

- **Monthly Trends (AreaChart):** 6-month historical curve visualizing reservation volume and attendee velocity over time.
- **Category Breakdown (Donut Chart):** Pie chart illustrating event distribution across Conference, Workshop, and Symposium categories.
- **Dynamic Pipeline Aggregation:** Backend MongoDB pipelines aggregate trends directly from live timestamps.
- **SVG Vector Rendering:** Lightweight, responsive charts featuring interactive hover tooltips and accessible color contrasts.
- **Zero Third-Party Trackers:** Client-side rendering preserves strict attendee privacy with zero external tracking scripts.

Administrative Reporting

Attendee Roster CSV Report

- **Dedicated Endpoint:** GET /api/reports/bookings/csv (Guarded by admin authorization).
- **Structured Fields:** Booking Reference, Attendee Name, Attendee Email, Event Title, Event Date, Seats Booked, Total Amount, Booking Date.
- **Operational Value:** Generates instant physical gate check-in rosters for admission verification and roll call.
- **Flattened JSON2CSV Stream:** Transforms populated MongoDB document hierarchies into clean flat tabular rows.
- **Streaming HTTP Response:** Streams text/csv with Content-Disposition attachment headers directly to browser.

Event Occupancy Summary CSV Report

- **Dedicated Endpoint:** GET /api/reports/events/csv (Guarded by admin authorization).
- **Structured Fields:** Event ID, Event Title, Category, Date, Venue, Total Capacity, Available Seats, Booked Seats, Occupancy Percentage, Status.
- **Operational Value:** Provides administrative audit records evaluating venue utilization and scheduling popularity.
- **RFC 4180 Compliance:** Strict adherence to comma-separated formatting standards, ensuring flawless parsing in Microsoft Excel, Google Sheets, and LibreOffice.
- **Instant Generation:** Executes in sub-100ms across all events without background batch delays.

Security Architecture &

1. Password Security & Hashing

Bcrypt password hashing with 10 salt rounds. Passwords are never stored in plain text and are excluded from query projections via Mongoose select: false.

2. Stateless JWT Authorization

Cryptographically signed HMAC-SHA256 bearer tokens with 7-day expiration. Role claims verified on all protected attendee and administrative endpoints.

3. NoSQL Injection Prevention

Strict Mongoose schema type-casting and object sanitization strip unauthorized MongoDB query operators (\$gt, \$where) from user inputs.

4. Audit Trail Preservation

Soft-cancellation safeguards prevent hard deletion of events with active bookings, eliminating orphaned records and preserving financial history.

5. Zero Credential Exposure

Environment isolation via .env files strictly excluded from source control. Automated git audits ensure zero secret exposure in public repositories.

UI Architecture & Screenshot Reference

CLEAN RESPONSIVE EXPERIENCE & SUBMISSION GUIDE

Application View	Route Path	Key Interface Components	Verified Functional Capability
Home Landing Page	/	Centered hero, value guarantees, capability cards, developer profile	Responsive value proposition with direct CTA buttons and developer bio
Events Catalog	/events	Search bar, category pills, real-time availability badges	Debounced keyword search and category filtering with instant updates
Event Details	/events/:id	Event schedule, venue, capacity progress bar, booking button	Detailed event information display with availability check
Ticket Booking Dialog	Modal Popup	Seat quantity selector (1-10), dynamic price tally	Concurrency-safe seat selection with instantaneous atomic reservation
Digital Ticket Pass	Modal Popup	Attendee name, reference code, QR stub, print view	Modal admission pass complete with printer-friendly styling
Attendee Dashboard	/dashboard	KPI counters, upcoming schedule list, pass preview	Personalized attendee portal with schedule tracking
My Bookings View	/my-bookings	Reservation history table, status badges, cancel button	Complete booking history with one-click cancellation rollback
Admin Dashboard	/admin/dashboard	KPI metric cards, Recharts Area & Donut charts, CSV buttons	Live telemetry console with one-click CSV export streaming
Manage Events	/admin/events	Event CRUD table, lifecycle toggles, safe cancellation	Organizer console for publishing, updating, and archiving events

Testing Methodology & Verification

77/77 AUTOMATED E2E TEST CASES PASSED

Section	Testing Domain	Test Cases	Validated Engineering Requirements & Edge Cases	Status
1	Health & Database	2 Cases	HTTP 200 health check, active MongoDB Atlas replica connection	100% PASS
2	Auth Edge Cases	11 Cases	Missing fields, weak password (<6), duplicate email, tampered JWT	100% PASS
3	RBAC Route Gates	12 Cases	Attendee 403 blocks on admin dashboard, user list, and CSV endpoints	100% PASS
4	Event Management	8 Cases	Public catalog list, search, category filter, negative capacity checks	100% PASS
5	Ticket Concurrency	13 Cases	0/negative seats, float values, >10 seat limits, draft event blocks	100% PASS
6	Booking Ownership	4 Cases	Owner pass view, reference code lookup, non-owner 403 blocks	100% PASS
7	Cancel & Rollback	5 Cases	Atomic seat restoration, double cancellation blocks, 404 handling	100% PASS
8	User Dashboard	3 Cases	Personal history list, cancelled count accuracy, upcoming metrics	100% PASS
9	Admin & CSV Exports	14 Cases	Dashboard KPIs, monthly trends, category pie, RFC 4180 headers	100% PASS
10	Safe Event Archival	5 Cases	Delete protection on booked events, cancellation transition, cleanup	100% PASS

Production Deployment

Multi-Tier Cloud Hosting Topology

- **Frontend SPA Hosting (Vercel):** React 19 single page application hosted on Vercel with vercel.json rewrite rules for seamless routing.
- **Backend REST API (Render):** Node.js / Express API hosted on Render PaaS with automated HTTPS and environment variable injection.
- **Database Tier (MongoDB Atlas):** Managed cloud replica set cluster featuring automated failover and IP access controls.
- **Live Frontend URL:** <https://skillorbit-event-booking-system.vercel.app/>
- **GitHub Repository:** <https://github.com/VenkataKarthikeya-eng/skillorbit-event-booking-system>

Build Optimization & Zero Secrets

- **Sub-Second Production Build:** Vite compiles 2,556 modules in ~845 milliseconds with zero compilation errors or syntax warnings.
- **Lightweight Bundle Footprint:** Optimized production distribution footprint (246 kB gzipped JS, 6.7 kB gzipped CSS).
- **CORS Security Binding:** Backend allows origins strictly matched to frontend deployment URL, blocking unauthorized cross-origin requests.
- **Strict .gitignore Auditing:** Multi-layer git exclusions guarantee that .env files, Mongo passwords, and JWT secrets are never committed.
- **Self-Healing Health Checks:** GET /api/health actively monitors database connection status and server uptime.

CURRENT BOUNDARIES & VERSION 2.0 SCOPE

Limitations & Future

Current Project Boundaries

- **Simulated Payment Flow:** Ticket pricing is modeled and calculated, but live credit card or UPI transactions are simulated without real money exchange.
- **Single-Timezone Scheduling:** Event schedules are currently anchored to campus local time without dynamic multi-timezone conversion.
- **Binary Role Hierarchy:** Role-based access distinguishes attendees from administrators without intermediate volunteer/gatekeeper roles.
- **In-App Notifications Only:** Immediate toast alerts and modal passes are provided; external email/SMS dispatch is not connected in current scope.
- **General Admission Model:** System enforces venue capacity limits but does not support reserved seat numbers in specific rows.

Version 2.0 Engineering Roadmap

- **Live Payment Gateway:** Integration of Stripe & Razorpay webhook pipelines with automated escrow handling and refund processing.
- **Encrypted QR Mobile Scanning:** Cryptographically signed QR codes for gatekeeper mobile check-in verification with offline validation.
- **Email & Calendar Dispatch:** SendGrid email integration delivering booking confirmations with attached .ics calendar invitations.
- **Interactive 2D/3D Seat Maps:** Visual SVG seating layouts allowing attendees to choose specific rows and seat numbers for auditoriums.
- **Tiered Organizer Permissions:** Multi-tenant organizer portal enabling department-level event ownership and revenue splitting.

Thank You

Capstone Delivery Summary

- ✓ **Architecture:** Full-stack decoupled MERN application with React 19 and Node.js v20.
- ✓ **Concurrency Control:** Atomic database-level updates guarantee zero double-booking.
- ✓ **Stateless Security:** JWT Bearer authentication and strict Role-Based Access Control.
- ✓ **Verified Quality:** 100% pass rate across 77 automated E2E integration test cases.
- ✓ **Live Deployment:** Production hosted on Vercel, Render, and MongoDB Atlas.

Evaluator Demo Credentials & Links

Administrator Demo Account:

Email: admin@skillorbit.com | Password: AdminPassword123!

Student Attendee Demo Account:

Email: student@skillorbit.com | Password: StudentPassword123!

Live Deployment URL:

<https://skillorbit-event-booking-system.vercel.app/>

GitHub Repository:

<https://github.com/VenkataKarthikeya-eng/skillorbit-event-booking-system>